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Energy Balance

$$\frac{\partial}{\partial t} \sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \quad (1)$$

Abbreviations:

w_i = engery densities (electric, magnetic, gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poyinting-Vectros $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

Mechanical

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Electromagnetic energy transfer

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (2)$$

$\varrho = \sigma \delta$ = charge density in interior

σ = charge density at border (surface)

δ = Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} \hat{=}$ elecectric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity = $|\bar{\mathcal{S}}|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (3)$$

Abbreviations:

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

J = "ignition of inhomogeneity, cause" j = Current density ϱ = Charge density

V = Potential difference I = Current R = Resistance

L = Inductance C = Capacitance Q = charge (ion) $\Psi = LI$ = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor}$ $r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$

$\rho = kr = \text{"radius of curvature } k\text{"}$

μ = permeability of conductor σ = conductivity of conductor (dielectricum)

Thermal

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Relativistic

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